Tier 1 Tier 2 Tier 3 Tier 4

(a) r = 0.86
RDF error [%]
CHGNet
MatterSim-v1.0.0-5M
UMA-S-P1 (compile)
UMA-S-P1 (turbo)
orb-v2
MACE-MP-0
orb-v3-direct-20-omat
GRACE-2L-OAM
GRACE-2L-MPtrj
orb-v3-conservative-inf-omat
UMA-M-P1 (turbo)
UMA-M-P1 (compile)
MACE-MPA-0
PET-OMAT-XL
NequIP-OAM-L
PET-OAM-XL
MACE-MH-1-OMAT
eSEN-30M-OAM
EquiformerV2

(b) r = -0.78
CHGNet
MatterSim-v1.0.0-5M
orb-v2
MACE-MP-0
GRACE-2L-OAM
GRACE-2L-MPtrj
MACE-MPA-0
NequIP-OAM-L
PET-OAM-XL
eSEN-30M-OAM
EquiformerV2

(c) r = 0.43
CHGNet
MACE-MP-0
GRACE-2L-OAM
orb-v2
MatterSim-v1.0.0-5M
GRACE-2L-MPtrj
MACE-MPA-0
NequIP-OAM-L
PET-OAM-XL
eSEN-30M-OAM
EquiformerV2

(d) r = 0.55
Pressure histogram error [%]
GRACE-2L-MPtrj
MACE-MP-0
MACE-MPA-0
CHGNet
orb-v3-direct-20-omat
EquiformerV2
MatterSim-v1.0.0-5M
orb-v3-conservative-inf-omat
PET-OAM-XL
orb-v2
NequIP-OAM-L
GRACE-2L-OAM
PET-OMAT-XL
UMA-S-P1 (turbo)
UMA-S-P1 (compile)
eSEN-30M-OAM
UMA-M-P1 (turbo)
UMA-M-P1 (compile)
MACE-MH-1-OMAT

(e) r = -0.77
CHGNet
GRACE-2L-MPtrj
MACE-MPA-0
MACE-MP-0
MatterSim-v1.0.0-5M
EquiformerV2
orb-v2
PET-OAM-XL
NequIP-OAM-L
GRACE-2L-OAM
eSEN-30M-OAM

(f) r = 0.45
CHGNet
GRACE-2L-MPtrj
MACE-MPA-0
MACE-MP-0
MatterSim-v1.0.0-5M
EquiformerV2
orb-v2
PET-OAM-XL
NequIP-OAM-L
GRACE-2L-OAM
eSEN-30M-OAM

(g) r = 0.79
VDOS error [%]
CHGNet
MACE-MP-0
GRACE-2L-MPtrj
orb-v2
MatterSim-v1.0.0-5M
NequIP-OAM-L
orb-v3-direct-20-omat
MACE-MPA-0
GRACE-2L-OAM
MACE-MH-1-OMAT
UMA-S-P1 (turbo)
orb-v3-conservative-inf-omat
eSEN-30M-OAM
UMA-S-P1 (compile)
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
EquiformerV2
PET-OAM-XL
PET-OMAT-XL
Force RMSE [eV/Å]

(h) r = -0.85
CHGNet
MACE-MP-0
MatterSim-v1.0.0-5M
orb-v2
GRACE-2L-MPtrj
NequIP-OAM-L
MACE-MPA-0
GRACE-2L-OAM
eSEN-30M-OAM
EquiformerV2
PET-OAM-XL
F1 score

(i) r = 0.42
CHGNet
MACE-MP-0
GRACE-2L-MPtrj
orb-v2
MatterSim-v1.0.0-5M
GRACE-2L-OAM
NequIP-OAM-L
MACE-MPA-0
eSEN-30M-OAM
EquiformerV2
PET-OAM-XL
κ_{SRME} score